Organic Chemistry

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ORGANIC CHEMISTRY

1. HYDROCARBONS
Alkanes: substitution (UV)

Markovnikov’s rule
Alkynes: addition, hydration

2. ALCOHOLS
Oxidation: 1 ->aldehyde->acid, 2 ->ketone
Dehydration: -> alkene
Esterification: ROH + RCOOH
Lucas, Iodoform tests

3. ALDEHYDES/KETONES
Nucleophilic addition
+HCN -> cyanohydrin
+Grignard -> alcohol
Reduction -> alcohol
Tollens: silver mirror
Fehling: red ppt
Iodoform: methyl ketones

4. CARBOXYLIC ACIDS
+ROH -> ester

Reduction -> 1 alcohol
Reactivity: AcylCl > Anhyd > Ester > Amide

5. AMINES
Basic
+Acid -> salt
Hinsberg, Carbylamine tests

6. AROMATIC
Electrophilic substitution:
Nitration, Halogenation, Sulfonation
Friedel-Crafts

7. STEREOCHEMISTRY

Chiral center: 4 different groups
Enantiomers, Diastereomers
R/S configuration
E/Z alkenes

8. MECHANISMS
SN1: 2 steps, carbocation
SN2: 1 step, inversion
E1, E2